Operating Systems Lab Project (40 Marks)

Project Guidelines

Objective

The laboratory project is intended to provide students with an opportunity to apply Operating Systems concepts to solve a practical problem through design, implementation, testing, and evaluation.

Students are expected to demonstrate their understanding of OS concepts by developing a working application, simulator, utility, or experimental framework using appropriate programming languages and tools.

Team Formation

- Students shall work in teams of maximum **three members**.
- Every member must contribute equally to the project.
- Individual contributions may be verified during project reviews and final evaluation.

Project Requirements

Projects should include implementation rather than only theoretical analysis. Projects should involve one or more Operating Systems concepts such as:

- CPU Scheduling
- Process Management
- Thread Programming
- Synchronization
- Deadlock Handling
- Memory Management
- Paging
- File Systems
- Virtual Memory
- Shell Programming
- Inter-Process Communication
- Process Monitoring
- Resource Allocation
- Linux System Programming
- Device Management
- Performance Analysis etc….

Deliverables

Each team shall submit:

- Project Proposal

- Source Code
- Documentation
- Test Cases and Results
- User Manual
- Final Presentation

Project Reviews

The project will be evaluated through **two review sessions** conducted during the final laboratory classes.

Review I

Students should demonstrate:

- Problem statement
- Objectives
- Literature/background study
- System design
- Architecture
- Algorithm/Flowchart
- Module division among team members
- Initial implementation (minimum working prototype)
- At least 60–70% implementation completed
- Core functionality should be executable

Review II

Students should explain the Operating Systems concepts used in their project. Students should demonstrate:

- Complete working implementation
- Testing
- Performance evaluation
- Error handling
- Demonstration of all features
- Documentation
- Viva voce

General Instructions

1. Each project must demonstrate one or more Operating Systems concepts.
2. Open-source libraries and frameworks may be used, but the core implementation must be developed by the team.
3. Proper code documentation and modular programming practices should be followed.
4. Any form of plagiarism copied code, or duplicate projects will result in disciplinary action as per the regulations.
5. Teams should maintain version history and backup of their work.

6. The project must execute successfully during the final demonstration.
7. All team members should be present during both reviews and the final evaluation.
8. The faculty may ask individual viva questions to assess each member's contribution.
9. Late submissions may attract penalties as per the course instructor's policy.
10. The decision of the course instructor regarding evaluation shall be final.

Examples of Suitable Project Topics

- CPU Scheduling Simulator with Comparative Analysis
- Virtual Memory and Paging Simulator
- Banker's Algorithm Resource Allocation Tool
- Process Scheduling Visualizer
- Deadlock Detection and Recovery Simulator
- Multithreaded File Search Utility
- Linux Process Monitor (Mini Task Manager)
- Mini Unix Shell with Built-in Commands
- Producer–Consumer and Reader–Writer Synchronization Simulator
- Memory Allocation Strategy Comparison (First Fit, Best Fit, Worst Fit)
- File System Analyzer and Disk Usage Monitor
- Inter-Process Communication Framework using Pipes and Shared Memory
- Thread Pool and Task Scheduling Framework
- Page Replacement Algorithm Simulator
- Mini Command-Line Backup and Restore Utility etc…

In VTOP, Project is divided into 4 fields, Every student should upload the document as per the following

Submission	Document to Upload	Description
Project (10 Marks)	**Project Proposal Report**	Problem statement, objectives, scope, literature/background, methodology, tools/software, work plan, and team member responsibilities.
Project (10 Marks)	**Project Design & Progress Report (Review-I)**	System architecture, module design, algorithms/flowcharts, database (if applicable), implementation progress (40–50%), screenshots, and challenges faced.
Project (10 Marks)	**Final Project Report**	Complete project documentation including introduction, objectives, design, implementation, testing, results, conclusion, future enhancements, references, and source code appendix (if required).
Project (10 Marks)	**Final Presentation (PPT)**	15-20 slides covering problem statement, objectives, architecture, implementation, demonstration, results, conclusion, and Q&A.

Project Collection form:

https://forms.gle/PGWzRso8xyV1TExV9